



Realize Your Career Dreams

GIFT School of Engineering and Applied Sciences

Spring 2019

CS-240: Object-oriented Programming

Lab-12 Manual

Interfaces and the ArrayList Class

Task #1: Comparable Interface Example

In this task, you are being asked to understand the use of Java **Comparable** interface.

In this task, you will learn about the **Comparable** interface of the standard Java library (**java.lang** package). The **Comparable** interface is used to compare two objects. The interface declares a single **compareTo** method that has the signature:

```
int compareTo(Object otherObject);
```

The call:

```
a.compareTo(b)
```

must return a *negative number* if **a** should come before **b**, *zero* if **a** and **b** are the *same*, and a *positive number* if **b** should come before **a**.

The **Comparable** interface has a single method:

```
public interface Comparable {  
    int compareTo(Object otherObject);  
} //interface
```

Now open and observe that the class **BankAccount** implements this interface and provides an implementation of the **compareTo** method that compares two account objects based on their balance. Study this class and understand how the **compareTo** method is written. Note that we must **cast** the **BankAccount** object from **Object** as in **line # 2** of this method.

Finally, open the **UsingBankAccounts** class where a test program is written for you that compares different **BankAccount** objects.

1. Open the **Lab Code** folder.
2. Open the **BankAccount** class. Understand the implementation of the **compareTo** method.
3. Open the **UsingBankAccounts** class. Note the array of **BankAccount** objects and how they are compared using the **compareTo** method.
4. Compile and run the code to see the output.

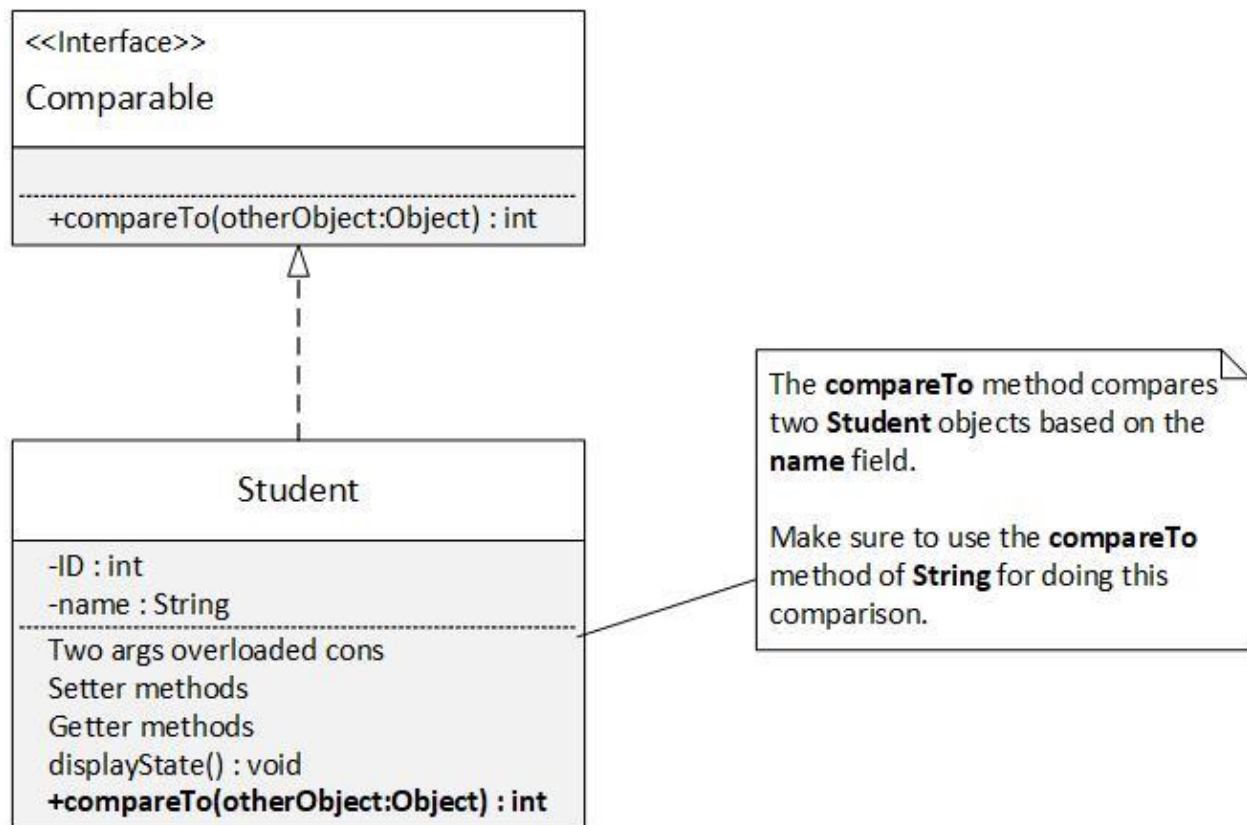
Task #2: Comparable Interface

In this task, you are being asked to implement the Java **Comparable** interface and override the **compareTo** method to understand how interfaces are used in Java.

NOTE: Write your classes and the *main* method in separate files.

DO NOT WRITE THE Comparable INTERFACE.

Convert the given UML diagram into classes. Note that **bold methods** represent **overridden methods** in the below given UML diagram.



1. Create all classes having names as per the above diagram, and **UsingStudents.java** having the **main** method.
2. Inside the **main** method, create **5 Student** objects using the overloaded constructor. Assign appropriate values to the state of all instances.
3. Display the state of all objects.
4. Demonstrate the use of **compareTo** method and compare these objects. Display the outputs using appropriate messages.

Task #3: ArrayList Class Example

In this task, you are being asked to understand the use of **ArrayList** class.

We continue to understand the usage of the **ArrayList** class (**java.util** package) as discussed in the lectures.

The **ArrayList** class is discussed in Chapter-7 Section 7.13 of your textbook.

Now, study the given **ArrayList** example paying attention to how to use its various methods, and write loops to display its contents.

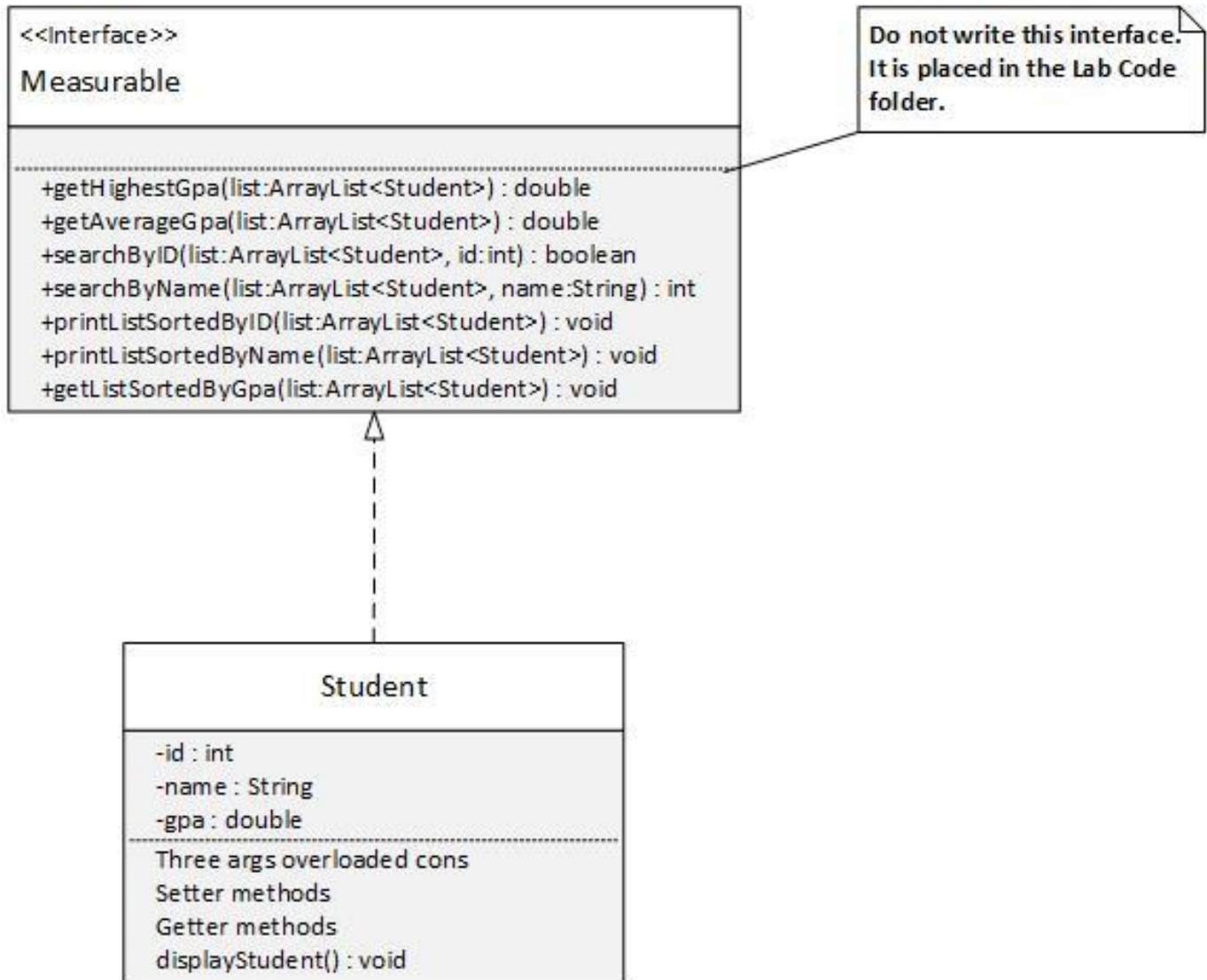
1. Open the **Lab Code** folder.
2. Open the **ArrayListLabDemo** class. Understand the use of various methods of this class.
3. Compile and run the code to see the output.

Task #4: ArrayList Class

In this task, you are being asked to use the **ArrayList** class and write methods that use this class.

NOTE: Write your classes and the *main* method in separate files.

Convert the given UML diagram into classes.



1. Open the **Lab Code** folder. Find the **Measurable** interface and the **Student** class in that folder.
2. Complete the implementation of the **Student** class.
3. Create an **UsingStudents** class having the **main** method.
4. Inside the **main** method, create an **ArrayList** of at least 5 **Student** objects using the overloaded constructor. Assign appropriate values to the state of all instances.
5. Display the state of all objects.
6. Demonstrate the use of **all methods** as given in the **Measurable** interface using appropriate messages.